Evidence of Decreased Heterodyne-Detection Efficiency Caused by Fast Beam Scanning in Wind Sensing Coherent Doppler Lidar, and Demonstration on Recovery of the Efficiency With Lag-Angle Compensation

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***Abstract*— The experimental evidence of the decreased heterodyne-detection efficiency caused by the lag angle is shown using a long-range wind sensing coherent Doppler lidar (CDL) with fast beam scanning. The recovery of the efficiency with the lag-angle compensation is also demonstrated. The receiving beam alignment method synchronized with the beam scanner is used. The measurable range of 12 km with the fast beam scanning of 20°/s in the case of 8-Hz line-of-sight (LOS) update rate was demonstrated with the compensation. This demonstration shows the potential for the wind sensing CDL to satisfy all requirements of long-range, real-time, and fast beam scanning measurement.**

***Index Terms*— Lag-angle compensation, lidar, scanning.**

I. Introduction

THE wind sensing coherent Doppler lidar (CDL) [1], [2] is a practical remote sensor for meteorological and aviation applications. In particular, the long-range (more than 10 km) CDL with the solid-state laser transmitter [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] is attractive for the wind monitoring around the airports [13] and the wide-area wind mapping for the efficient wind energy production [14]. For these applications, fast beam scanning is required for real-time wind mapping in addition to the long-range wind measurement.

The abovementioned existing CDLs [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] already have the almost theoretically limited system efficiency. This means the CDLs have had: 1) nearly diffraction-limited beam quality; 2) nearly shot-noise-limited optical-electrical conversion; and 3) almost aberration-free optical components with small optical losses. However, there is one remaining term regarding the system efficiency, which might have been decreased in the past CDLs. This is the heterodyne-detection efficiency caused by the "lag angle" in the case of fast beam scanning. The issue on the abovementioned lag angle has been described conceptually in the past papers [15], [16]. The necessity of the lag-angle compensator has been pointed out, especially for the spaceborne CDL [16]. However, to the best of our knowledge, the experimental evidence of decreased heterodyne-detection efficiency caused by the lag angle has not been seen in the past literature. If the effect of the lag-angle compensation is demonstrated, this can be the last big progress in the performance of the wind sensing CDL regarding the system efficiency.

Here, we show the evidence of the decreased heterodyne-detection efficiency caused by the lag angle in the case of fast beam scanning. Our long-range wind sensing CDL technology [9], [10] is used for showing the abovementioned evidence. Furthermore, we also demonstrate on the recovery of the efficiency with the lag-angle compensation. In this compensation, we apply and modify the technology of the receiving beam alignment which we recently reported [17]. In the following, the influence of the lag angle on the performance of CDL is theoretically explained in Section II. The system configuration of the CDL with the lag-angle compensator is shown in Section III. The method of lag-angle compensation is also explained. Then, the experimental evidence of the decreased heterodyne-detection efficiency caused by the lag angle is shown, and the recovery of the efficiency with the lag-angle compensation is demonstrated in Section IV. The measurable range of 12 km with the fast beam scanning of 20°/s in the case of 8-Hz line-of-sight (LOS) update rate is demonstrated. Some parts of this article have been reported in the related conference papers [18], but more detailed and new explanations, investigations, and results are shown here.

II. Influence of Lag Angle in Performance of CDL

The wind sensing CDL normally has the coaxial optics to maximize the system efficiency. When the transmitting and receiving beam is scanned, the lag angle appears between the two beams. The schematic of the lag angle is shown in Fig. 1. The principle of the constancy of the speed of light tends to make us to misunderstand that the influence of the lag angle is negligible. However, it is a little bit surprising that

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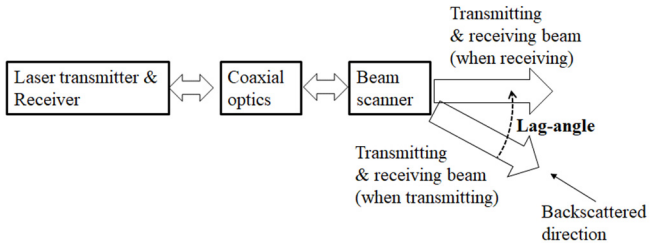


Fig. 1. Schematic of lag angle.

the influence exactly exists in the performance of the wind sensing CDL. This influence is explained theoretically in the following.

The lag angle in the case of coaxial optics at the scanner output (rad) is expressed by

$$\delta(V_S, L) = \frac{2 \cdot \frac{\pi \cdot V_S}{180} \cdot L}{c} \tag{1}$$

where c is the speed of light (m/s), L is the measurement range (m), and V_S is the scanning speed (°/s). The field of view (FOV) of the CDL is approximately equivalent to the $1/e^2$ divergence (full width) of the receiving beam (rad) and is expressed as

$$\theta_D \approx \frac{4\lambda}{\pi D_b} \tag{2}$$

where λ is the laser wavelength (m) and D_b is the $1/e^2$ intensity beam diameter (m). When the lag angle is larger than the FOV, the receiving efficiency (i.e., heterodyne-detection efficiency) becomes decreased. For example, under the condition of the scanning speed: 20°/s, measurement range: 10 km, wavelength: 1.55 μm, and beam diameter: 88 mm, the lag angle at the receiving aperture becomes 23 μrad and larger than the FOV of 22 μrad. In such a case, the decreased heterodyne-detection efficiency appears distinctly. Equations (1) and (2) show that the larger receiving beam diameter makes the influence of the lag angle more severe, even though it is generally better for the long-range measurement in the case of without scanning.

The heterodyne-detection efficiency is quantitively calculated as [19], [20]

$$\begin{aligned}
&\eta_C(\delta(V_S, L), \lambda, D_b, D, L, F) \\
&= \frac{256\beta^2\varsigma^4}{\pi\left[1-\exp\left(-\varsigma^2\right)\right]^2} \int_0^\infty \frac{\rho}{D} d\left(\frac{\rho}{D}\right) \int_0^{2\pi} d\theta \\
&\quad \times \left| \int_0^{\frac{1}{2}} r dr \exp\left[-2\varsigma^2 r^2 + \frac{j}{2}(\beta-\alpha)r^2\right] \times J_0\left(\frac{\beta\rho r}{D}\right) \right|^2 \\
&\quad \times \left| \int_0^{\frac{1}{2}} r dr \exp\left[-2\varsigma^2 r^2 + \frac{j}{2}(\beta-\alpha)r^2\right] \right. \\
&\qquad \left. \times J_0\left\{ r\left[\beta^2\left(\frac{\rho}{D}\right)^2 + \xi^2 - 2\beta\left(\frac{\rho}{D}\right)\xi\cos\theta\right]^{\frac{1}{2}} \right\} \right|^2
\end{aligned} \tag{3}$$

where J_0 is the zeroth-order Bessel function, D is the receiving aperture diameter (m), r is the distance from the center at the receiving aperture plane (m), and ρ is the distance from the beam center at the target range plane (m). In the equation, α, β, ς, and ξ are the values expressed by the following equations to make the argument for exponential in (3) dimensionless

$$\alpha = \frac{kD^2}{F} \tag{4}$$

$$\beta = \frac{kD^2}{L} \tag{5}$$

$$\varsigma = \frac{\sqrt{2}D}{D_b} \tag{6}$$

$$\xi = kD\delta(V_S, L) \tag{7}$$

where F is the beam focusing distance (m) and k is the wavenumber ($=2\pi/\lambda$).

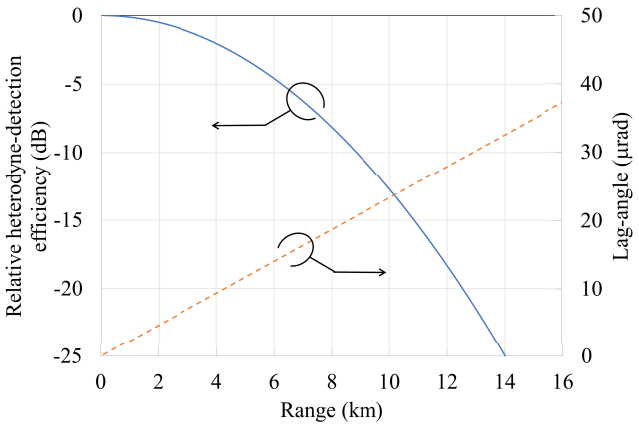


Fig. 2. Example of calculation result of heterodyne-detection efficiency (normalized by the efficiency without scanning) and lag angle.

TABLE I
PARAMETERS FOR CALCULATION OF HETERODYNE-DETECTION EFFICIENCY

Parameter	Value
Laser wavelength	1.55 μm
Beam diameter	88 mm
Focal range	∞
Receiving aperture diameter	110 mm
Scanning speed	20 deg/s

The example of the calculated result using (3) regarding the range dependence of heterodyne-detection efficiency is shown in Fig. 2. The second vertical axis shows the lag angle at the scanner output. The parameters for the calculation are shown in Table I. The values in the table correspond to the CDL, which we use in the demonstration of the later section. It is shown that the decrease in the efficiency becomes distinct at the range of longer than 5 km. This also implies that there is a possibility in the big progress of the system efficiency in the case of the long-range measurement with fast beam scanning if the impact of the lag angle can be compensated.

The alternative idea to prevent the impact of the lag angle is the deployment of digital stepped scanning. The lag angle does not appear for this scanning in principle. However, we use analog scanning with the lag-angle compensation since the digital stepped (stop and go) scanning damages the motor for the scanner and makes its lifetime short, especially in the case of the large scanner.

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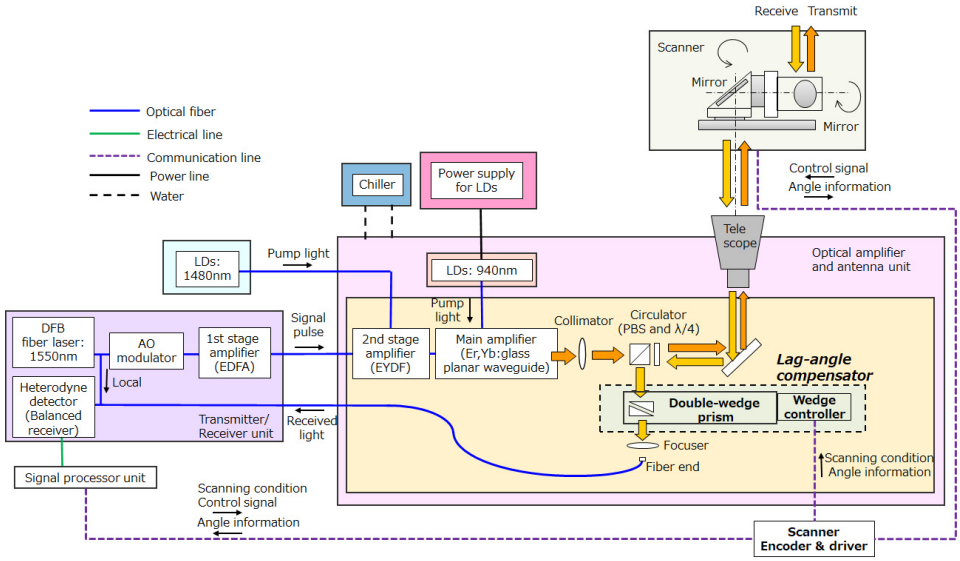


Fig. 3. Schematic of wind sensing CDL with lag-angle compensator.

III. System Configuration of Wind Sensing CDL With Lag-Angle Compensator

A. System Configuration

The system configuration of the wind sensing CDL with the lag-angle compensator is shown in Fig. 3. The appearance of the CDL is shown in Fig. 4. The basic concept is the combination of the existing 1.55-μm long-range CDL [9], [10] and a newly installed lag-angle compensator. The CDL has an optical transmitter/receiver unit [21], a main amplifier unit, 1480-nm laser diodes (LDs), a chiller, a power supply for the LDs, a beam scanner, a scanner control unit, and a signal processor unit.

The optical transmitter/receiver unit provides the basic functions of the wind sensing CDL and has a 1.55-μm distributed feedback (DFB) fiber laser, a balanced receiver, an acousto-optic modulator (AOM), and an erbium-doped fiber amplifier (EDFA). The EDFA provides the first stage of optical amplification.

The optical amplifier and antenna unit contains a second-stage amplifier, a main amplifier, pumping LDs at 940 nm, a telescope, a circulator, and a lag-angle compensator. The second-stage amplifier is the erbium–ytterbium-doped fiber pumped by the 1.48-μm LDs, which is described in [22]. The main amplifier is the erbium–ytterbium-doped planar waveguide amplifier [9], [10]. The circulator consists of a polarization beam splitter (PBS) and a 1/4-wave plate. The lag-angle compensator consists of a double wedge prism synchronized with the scanner controller. The wedge prism is the antireflection coated fused silica. The effective receiving aperture of the telescope is 110 mm with the transmitter/receiver beam diameter of 88 mm. The beam scanner has the function of hemispherical scanning using two mirrors.

B. System Operation

The basic operation for wind sensing is the following. The output of the DFB fiber laser is divided into a transmitting light and a local oscillator light for heterodyne detection. The transmitting light is pulse-modulated and frequency-shifted by the AOM. The optical pulses are amplified to a peak power of about 5 W at the first stage amplifier. The second-stage amplifier increases the peak power to a few tens of watts. The main amplifier is operated with a peak power of about 2 kW and a pulse duration of about 500 ns (i.e., a pulse energy of about 1 mJ). The pulse repetition frequency is 4 kHz. The output of the main amplifier is transmitted to the atmosphere through a circulator and a telescope. The transmitted light is backscattered by the aerosol particles in the atmosphere. The backscattered light is passed through the optics, coupled to an optical fiber, and sent to the balanced receiver for heterodyne detection. The detected signals are processed in the signal processor unit. In signal processing, the signals are gated by the timed gates corresponding to the measurement ranges. The spectra for the ranges are calculated by the fast Fourier transform. The incoherent accumulation is performed to improve the detection probability and the velocity estimation accuracy. The noise floor reductions are performed for the accumulated spectra in the postprocessing. The LOS velocities of the ranges are estimated from the first moments of the postprocessed spectra around their peak frequencies. The peak frequency search range corresponds to the LOS velocity search range (in the experiments of this article, from -30 to $+30$ m/s). The range dependence of the



(a) (b)

Fig. 4. Appearance of CDL. (a) Overall view. (b) Scanner.

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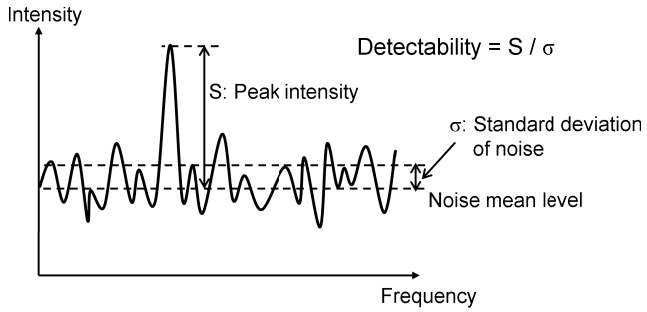


Fig. 5. Definition of detectability.

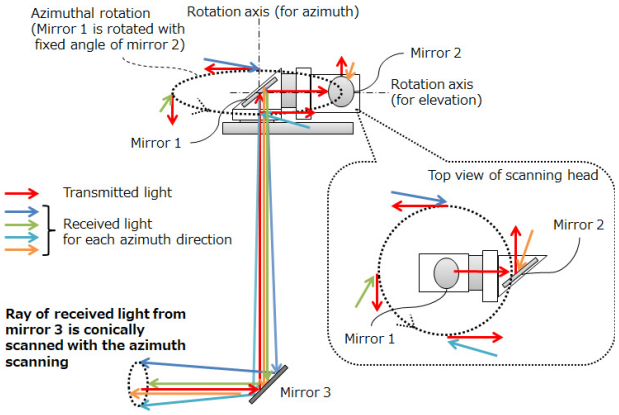


Fig. 6. Schematic of lag angle in the case of PPI scanning.

detectability is also calculated using the accumulated spectra. Note that the detectability is equivalent to the spectral domain signal-to-noise ratio (SNR) and proportional to the heterodyne-detection efficiency. The definition of detectability is shown in Fig. 5 [10] [17]. The beam scanning with the lag-angle compensation is performed in parallel with the abovementioned LOS velocity measurement. Then, the distribution of the LOS wind velocity and detectability is visualized.

The total system efficiency of the CDL without beam scanning is about 10%. This efficiency includes: 1) the theoretical limit of the heterodyne detection (40% [23]); 2) the quantum efficiency of 80% of the balanced receiver; 3) the influence of the beam quality; and 4) the insertion loss of the optical components. The decrease of the system efficiency caused by the lag angle is shown in Fig. 2 since the relating parameters are the same as the ones for the figure. The operation of the lag-angle compensation is explained in Section III-C.

C. Lag-Angle Compensation

The lag-angle compensation is performed by the beam alignment of the receiving beam using the double wedge prism. This concept of the alignment of the receiving beam is the same as the one which we recently reported [17].

Fig. 6 shows the schematic of the lag angle at the double wedge prism of Fig. 3 in the case of the plan position indicator (PPI). The lag angle can be compensated if the angle difference between the transmitting and receiving beams is controlled so as to match with the lag angle. This control is easy if both the value and direction of the compensation at the double wedge prism are constant regardless of the measurement direction. However, it is known from Fig. 6 that the direction of the lag angle at the double wedge prism unfortunately changes depending on the measurement direction. This makes the operation in the lag-angle compensation complex. Furthermore, the lag angle also depends on the elevation measurement direction and is expressed as

$$\theta_{\mathrm{LA}}^{P}\left(\theta_{\mathrm{EL}}^{P}\right) = a\cos\left[\sin^2\left(\frac{\pi}{2} - \theta_{\mathrm{EL}}^{P}\right)\left(\cos\delta(V_S, L) - 1\right) + 1\right] \quad (8)$$

where θ_{EL}^{P} is the elevation angle in the case of PPI scanning (rad). This means that the lag-angle compensation must be synchronized with the scanner controller. In our demonstration on the lag-angle compensation at the later section, we use PPI scanning. Therefore, we have installed the synchronization algorithm into the wedge controller. For the case of the range height indicator (RHI) scanning, mirror 2 in Fig. 6 is rotated by fixing mirror 1. The ray of the received light from mirror 3 is conically scanned as well as the case of PPI scanning in Fig. 6. The condition of the RHI scanning is equivalent to the horizontal scanning (with the elevation angle of 0°) inclined to the orthogonal direction. Consequently, the lag angle for the RHI scanning is calculated by assuming $\theta_{\mathrm{EL}}^{P} = 0$ in (8).

The compensation operation acts as follows. The scanning speed and the target range for the compensation are set before the measurement. Then, the compensation angle and the speed of the wedge rotation are calculated with (8), and the double wedge is set to make the compensation angle. After that, the beam scanning and LOS wind velocity measurement start. This triggers the rotation of the double wedge by keeping their relative position to each other. The rotation speed of the set of double wedge prism corresponds to the scanning speed.

The beam alignment precision at the scanner output has been designed to be smaller than 1 μrad. This is achieved by designing: 1) the wedge angle; 2) the refractive index of the wedge; 3) the motor rotation precision; 4) the gear precision of the small and large gear; and 5) the gear reduction ratio. It should be noted that the lag-angle compensation is performed at the double wedge prism.

A certain distance exists between the prism and the scanner output. This caused the difference regarding the beam center position between the transmitting and the receiving beam. However, the impact of this mismatching of the beam centers does not impact the heterodyne-detection efficiency due to the following investigation. The compensation angle is the order of a few tens of microradians and the distance between the prism and scanner output of CDL is about 1 m. This means that the difference in the beam center position is about a few tens of micrometers. This difference is much smaller than the beam diameter (i.e., 88 mm) and the impact on the heterodyne-detection efficiency is negligible.

The additional lag angle is caused by the synchronization angle error between the scanning direction and the rotational angle of the set of double wedge prism. This is approximately given by

$$\sigma \approx \frac{\phi\delta_M}{N_T} \quad (9)$$

where ϕ is the synchronization angle error (rad), N_T is the magnitude of telescope, and δ_M is the maximum angle of

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TABLE II
EXPERIMENTAL CONDITION OF CDL

Parameter	Value
Laser wavelength	1.55 μm
Pulse energy	1 mJ
Pulse width	500 ns
Pulse repetition frequency	4 kHz
Beam diameter	88 mm
Beam focusing distance	∞
Receiving aperture diameter	110 mm
Range resolution (range gate width)	150 m
Receiving bandwidth (corresponding to range gate width)	1 MHz
LOS velocity update rate	8 Hz
Beam scanning speed	1 deg/s, 20 deg/s
Beam elevation angle	1.5 deg
Beam azimuth scanning width	180 deg
Target range for lag-angle compensation	10 km
System efficiency (roughly estimated)	-10 dB
Lidar location altitude	20 m

compensation by wedge prisms (rad). The designed value of σ is 0.6 μrad with the values of ϕ: 1.7 mrad, N_T: 22, and δ_M: 7.7 mrad.

Our approach using double wedge prisms is the simpler method compared with the compensator using a mirror. If the lag angle is compensated with a single mirror instead of the double wedge prism, the complex biaxial tilt like a conical scan must be needed to match the measurement direction. If the compensator consists of two mirrors, the degree of freedom becomes 4. This makes the synchronized control even more complicated. In the approach of the double wedge prism, the relative rotational angle between two wedges can be fixed if the scanning speed is constant. This means that the double wedge approach is simple with only one degree of freedom (i.e., the rotational angle). The nonmechanical scanning device, such as liquid crystal polarization gratings (LCPGs) [24], can be the other candidate. However, this device is in the development step for the CDL application and is not available at this moment.

IV. EXPERIMENTAL EVIDENCE OF IMPACT OF LAG ANGLE AND RECOVERY OF HETERODYNE-DETECTION EFFICIENCY WITH LAG-ANGLE COMPENSATION

A. Experimental Condition and Procedure

An experiment to see the evidence of the impact of lag angle is introduced here. Furthermore, the recovery of the heterodyne-detection efficiency with the lag-angle compensation is demonstrated. The experiment was performed on the roof of a building adjacent to the Sendai Airport.

The experimental parameters for the CDL measurement are shown in Table II. The beam scanning is the PPI with the elevation angle of 1.5°. The target range for the lag-angle compensation is set at 10 km since the data availability at this range is empirically important for the application to airport wind monitoring. The scanning speed is set at 1°/s and 20°/s. In the case of the scanning speed of 1°/s, the decrease of the heterodyne-detection efficiency is obtained using (3) and is less than 0.1 dB at the target range of 10 km. This means that the difference in the detectability between the two scanning speed cases shows the decrease in the heterodyne-detection efficiency. The LOS update rate is 8 Hz, which corresponds to the spectral accumulation number of 500. This implies the measurement angle resolution of 0.125° and 2.5° for the scanning speed of 1°/s and 20°/s, respectively. The azimuth scanning width is limited to 180° to prevent from the shadowed area by the buildings.

TABLE III
PROCEDURE OF EXPERIMENT

Step	Scanning speed	Lag-angle compensation	Remarks
1	1 deg/s	Without compensation	For checking the performance without influence of lag-angle
2	20 deg/s	Without compensation	Steps 2 and 3 are iterated for 7 times alternately
3	20 deg/s	With compensation	
4	1 deg/s	Without compensation	For confirmation of repeatability of step 1

The procedure of the experiment is given as follows. In step 1, the LOS wind velocity and detectability distribution are measured with PPI scanning for 1°/s without lag-angle compensation. In step 2, the measurement is performed with the scanning speed of 20°/s without lag-angle compensation. In step 3, the same measurement but with the compensation is performed. Steps 2 and 3 are iterated seven times to confirm the repeatability of the results. In step 4, the repeatability regarding step 1 is confirmed. The contents of these steps are shown in Table III.

B. Theoretical Prediction of Range Dependence of Detectability

Before the experiment, the range dependence of detectability has been roughly predicted theoretically. It is calculated as

$$D_N(L) = \frac{\eta_d(L) \cdot \eta_O \cdot \lambda \cdot E \cdot \beta \cdot exp(-2\alpha_E L) \cdot \pi \cdot D^2}{8 \cdot h \cdot B \cdot L^2} \cdot \sqrt{N} \tag{10}$$

where E is the pulse energy (J), β is the aerosol backscattering coefficient (m/sr), α_E is the atmospheric extinction coefficient (km), h is Plank's constant (Js), B is the receiving bandwidth (Hz), N is the spectral accumulation number, and η_O is the optical system efficiency. The heterodyne-detection efficiency, $\eta_d(L)$, is given by

$$\eta_d(L) = \frac{\eta_F}{1 + \left(1 - \frac{L}{F}\right)^2 \cdot \left(\frac{\pi (AD)^2}{4\lambda L}\right)^2 + \left(\frac{AD}{2S_0(L)}\right)^2} \cdot \eta_S(L) \tag{11}$$

where η_F is the far-field heterodyne-detection efficiency and A is the correction factor describing the beam truncation of the telescope when the nearest Gaussian approximation is adopted. The value of this correction factor is 0.71 in the

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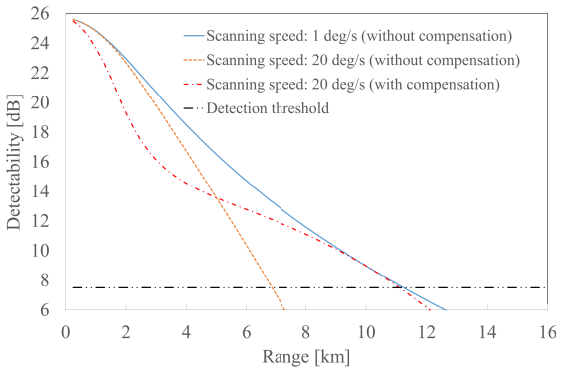


Fig. 7. Calculated results of range dependence of detectability with assumed atmospheric parameters for each scanning condition.

optimized beam truncation condition of $D_b/D = 0.8$. The value of η_F at this condition is 0.4. This equation is based on the SNR calculation [23] with additionally including the effect of the incoherent spectral accumulation. $S_0(L)$ is the transverse coherent length (m) and can be written as

$$S_0(L) = \left[2.91k^2 \int_0^L C_n^2(z)\left(1 - \frac{R}{L}\right)^{5/3} dR\right]^{-\frac{3}{5}} \tag{12}$$

where $C_n^2(z)$ is the refractive index structure constant ($\mathrm{m}^{-2/3}$) and z is the altitude of the range (m) and given by $R \cdot \sin\theta_{\mathrm{EL}}$ [θ_{EL}: beam elevation angle (rad)]. $\eta_S(L)$ is the efficiency, which expresses the decrease of the heterodyne-detection efficiency caused by the lag angle, and this is given by

$$\eta_S(L) = \frac{\eta_C(\delta, \lambda, D_b, D, L, F)}{\eta_C(0, \lambda, D_b, D, L, F)}. \tag{13}$$

Note that this is the ratio of the two efficiencies with and without scanning.

The range dependence of detectability is calculated using (10) and the parameters are shown in Tables I–III. The aerosol backscattering coefficient and the atmospheric extinction coefficient were roughly assumed at the experiment site using a particle counter and were estimated as 1.4×10^{-7} m/sr and 0.04 /km. These two atmospheric parameters were assumed to be constant since the wind sensing experiments were performed within the planetary boundary layer. The submarine laser communication (SLC) night model [25] was assumed as the refractive index structure constant (see the Appendix). The calculated results for each scanning scheme of Table III are shown in Fig. 7. The detectability of the spectral noise peak is empirically lower than 7.5 dB in our wind sensing CDL measurement, and we define this value as the signal detection threshold. This detection threshold is shown in Fig. 7 and is utilized for the evaluation of the measurable range. It is known that the detectability is decreased by the scanning with 20°/s, especially for the far range, but it is recovered in the case of the lag-angle compensation.

Fig. 7 also shows that the effect of the compensation is negative in the near ranges (in the figure, <5 km). This is reasonable since the compensated angle does not match these ranges. In other words, the compensation makes the lag angle larger, especially for the short ranges. It is ideal if the lag-angle compensation is performed with different compensation angles

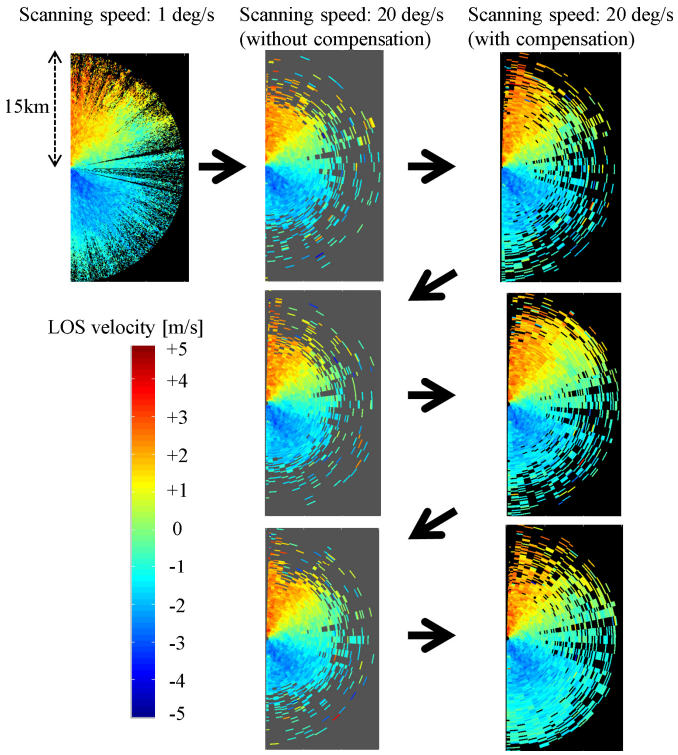


Fig. 8. LOS wind velocity map with PPI scanning for step 1 and the first three iterations of steps 2 and 3 in Table III (arrows show the order of the measurements).

depending on the ranges for a single pulse shot. However, it requires the lag-angle compensator with very high response time. For example, if the compensation needs to be done for all ranges from 0 to 15 km, the compensation must be done within 200 μs. The ideal compensation may be realistic if the nonmechanical scanning device such as LCPGs [24] is used. Unfortunately, this device is not available at this moment as described in Section III-C. On the other hand, the method of the double wedge prism enables the lag-angle compensation ideally only at the target range, but the method exactly enhances the measurable range by setting the suitable target range (see Fig. 7). If the long measurement interval is acceptable, the compensation can be done for all ranges by changing the target range little by little with changing the relative rotational angle between two wedges.

It should be noted that the results in Fig. 7 have been calculated just to imply the effect of the lag-angle compensation schematically using one example of the atmospheric parameters. Therefore, we use the result of this figure as the reference for the investigation of the later-shown experimental results.

C. Experimental Results

Figs. 8 and 9 show the maps of the LOS wind velocity and detectability for step 1 and the first three iterations of steps 2 and 3. It is visually shown that the measurable range was decreased by changing the scanning speed from 1°/s to 20°/s and the measurable range was recovered with the lag-angle compensation. The detectability is quite low for some azimuth directions. This is mainly because of the distribution

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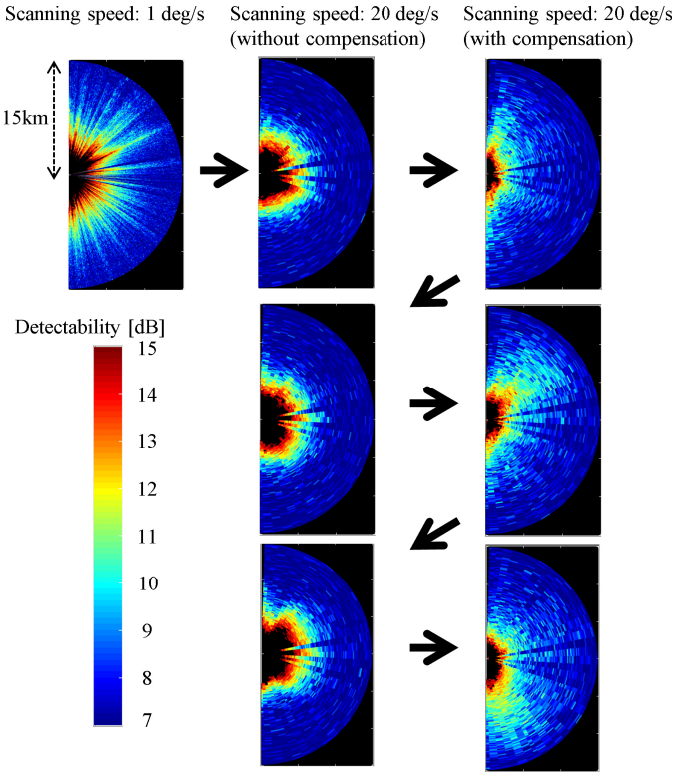


Fig. 9. Detectability map with PPI scanning for step 1 and the first three iterations of steps 2 and 3 in Table III (arrows show the order of measurements).

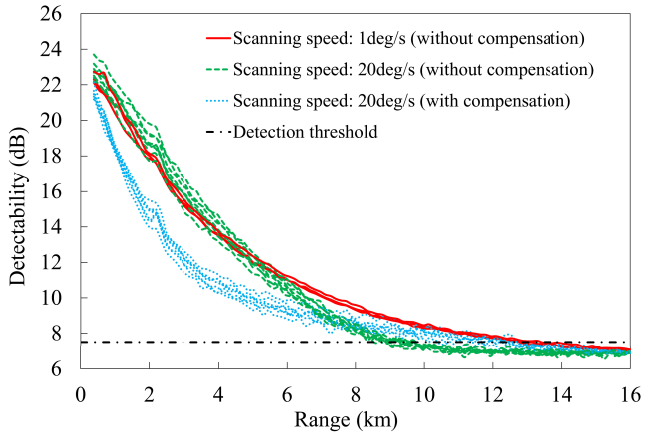


Fig. 10. Range dependence of detectability with and without lag-angle compensation.

regarding the aerosol backscattering coefficient. The slight blocking by the buildings might have occurred for the azimuth angles that have no data.

Fig. 10 shows the range dependence of the detectability averaged for all the measurement directions (i.e., for the azimuth angle width of 180°). Each line corresponds to each iteration number. The results for the abovementioned steps of 1–4 are plotted. The lines asymptotically approach about 7 dB at long range. This corresponds to the random outliers of the peak frequency detection in the spectra of the received signals, which is not considered in the theoretical calculation of Fig. 7. It is obvious that the repeatability can be seen in the decrease of detectability in the case of high scanning speed. This is

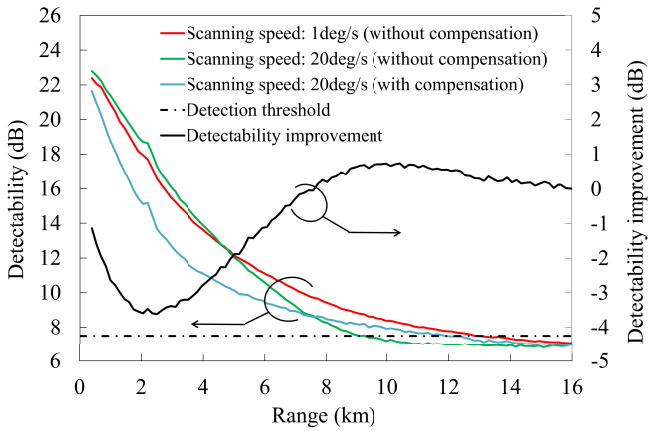


Fig. 11. Range dependence of detectability with and without lag-angle compensation. The lines for each scanning condition are averaged, and detectability improvement by lag-angle compensation is added.

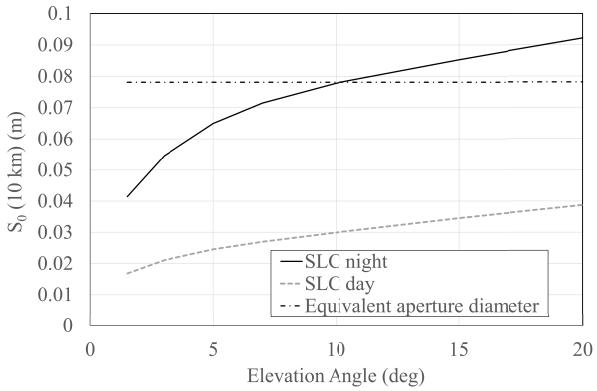


Fig. 12. Simulated transverse coherence length at a range of 10 km versus elevation angle.

the evidence of the decreased heterodyne-detection efficiency caused by the lag angle. The recovery of the efficiency by the lag-angle compensation is also sure, especially around the target range for the compensation (i.e., 10 km).

The range dependence of detectability is additionally averaged for all the iteration numbers regarding each step of 1–3. The results are plotted in Fig. 11. The second axis shows the difference of detectability between the cases with and without lag-angle compensation (i.e., between steps 2 and 3). The maximum recovery of detectability (i.e., the recovery of heterodyne-detection efficiency) is 0.7 dB, and it appears at the range of 10 km, which agrees with the designed value. Furthermore, it is obvious that the lag-angle compensation decreased the heterodyne-detection efficiency at the near ranges. The increased efficiency is also obvious at the far ranges around the target range for the compensation. This tendency of the range dependence of detectability with and without lag-angle compensation agrees qualitatively well with the calculated result of Fig. 7.

On the other hand, although the theory predicts the improvement of more than 10 dB (see Fig. 2), the experiments show 0.7 dB as abovementioned. This discrepancy did not result from the insufficient effect of the lag-angle compensation. In fact, the detectability of the target range (i.e., 10 km) recovered almost perfectly by the lag-angle compensation

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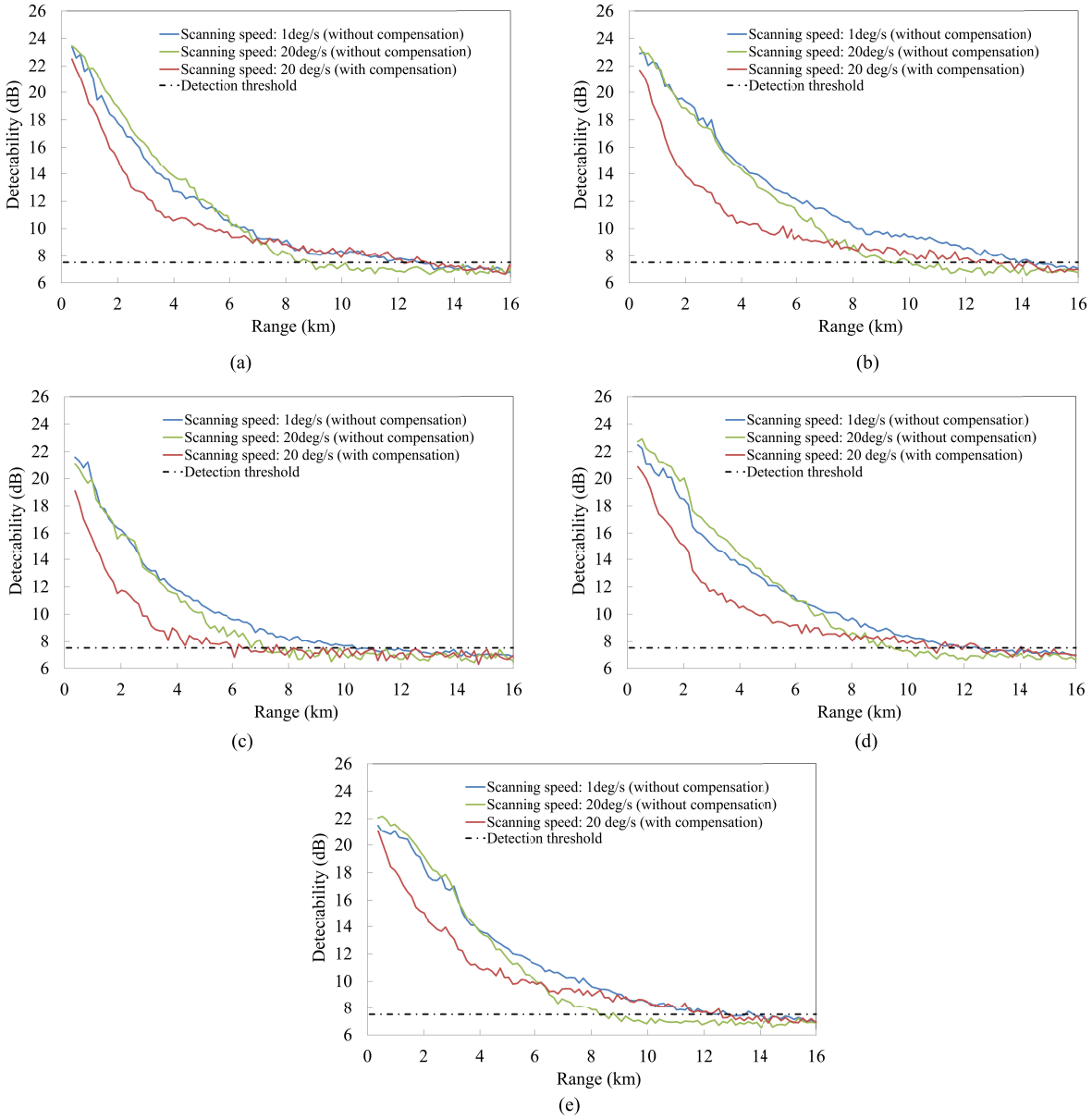


Fig. 13. Range dependence of detectability with and without lag-angle compensation for each azimuth direction region. (a) 20°–30°. (b) 60°–70°. (c) 78°. (d) 110°–120°. (e) 150°–160°.

(see Fig. 11). The reason for the discrepancy is that the decrease of the heterodyne-detection efficiency in the case of high scanning speed was much smaller than the theoretical value. The exact reason for this smaller decrease is not sure, but the aberration could be the candidate. The aberration makes the transmitting and receiving beams diverge and reduces the negative influence of the lag angle. Note that the theory [see (10)–(13)] includes the influence of the aberration and lag angle independently. This means that the mixed effect of these influences does not appear in the calculated results. Therefore, this mixed effect is additionally investigated as follows.

The aberration could be caused by the refractive turbulence or the wavefront distortion of the optics. The former is considered as the main source since the latter has been designed carefully with reduced aberration. If this investigation is valid, the effect of the lag-angle compensation will be more distinct in the measurement under small refractive turbulence. The refractive turbulence typically becomes small under the clear and calm night around the sunset and at higher altitudes. The measurement with larger elevation angle reduces the integrated influence of the turbulence along the path. In such cases, the decrease of the heterodyne-detection efficiency by the lag angle becomes large, and more distinct effect regarding the lag-angle compensation could potentially be observed. This is the mixed effect of the influences of the lag angle and the optical aberration.

Fig. 12 shows the simulation result of the elevation angle versus the transverse coherent length at the target range of 10 km. The SLC day model [25] (see the Appendix) is additionally assumed in addition to the night model, which was used in the calculation of Fig. 7. The equivalent aperture diameter [$=A_c \times D$ in (11)] is also shown for the reference. Theoretically, the influence of the turbulence becomes distinct

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TABLE IV

AVERAGE MEASURABLE RANGES FOR EACH AZIMUTH DIRECTION REGION (UNIT: km)

Direction \ Scanning Speed	1 deg/s	20 deg/s (without compensation)	20 deg/s (with compensation)
0 deg ~ 180 deg	12.8	9.4	12.1
20 deg ~ 30 deg	13.0	9.1	12.2
60 deg ~ 70 deg	14.8	9.5	13.1
78 deg	10.4	7.1	6.1
110 deg ~ 120 deg	12.7	9.5	11.0
150 deg ~ 160 deg	13.0	8.3	12.7

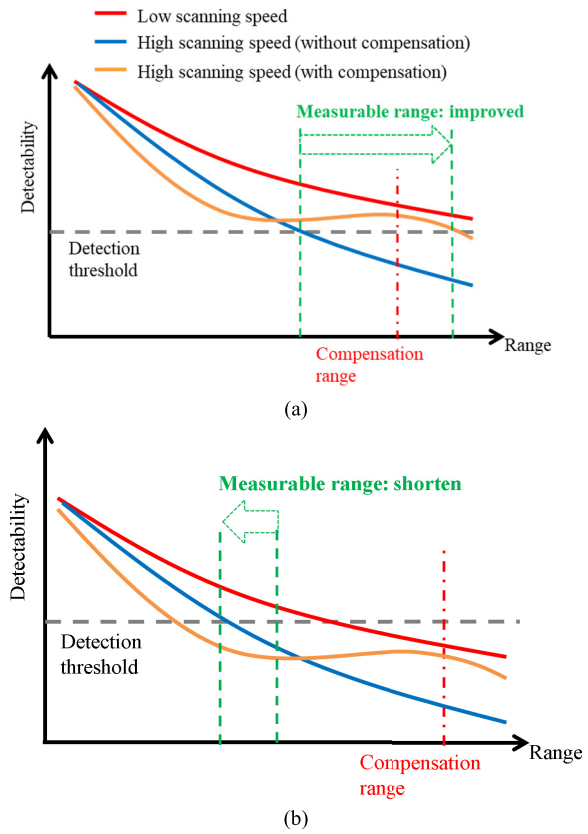


Fig. 14. Schematic of range dependence of detectability in (a) high and (b) low detectability cases.

when the transverse coherence length $[S_0(L)]$ becomes shorter than the equivalent aperture diameter [see (11)]. In this condition, the decrease of the heterodyne-detection efficiency becomes smaller than the calculated value. The experiment with the elevation angle of 1.5° is under this condition (see Fig. 12).

The range dependences of averaged detectability for some azimuth measurement direction regions are shown in Fig. 13. The measurable range for each azimuth direction region is summarized in Table IV. The measurable range is defined as the shortest range of which the averaged detectability for the azimuth direction region is lower than the detection threshold of 7.5 dB. The recovery of detectability and measurable range is obvious other than in the case of the specific azimuth angle of 78°. The average measurable range for the directions other than this specific azimuth angle is 12.2 km. To the best of our knowledge, this is the longest wind sensing demonstration with such a fast scanning speed (i.e., 20°/s) and high LOS velocity update rate of 8 Hz, although there were many wind sensing results of more longer measurable range but with lower scanning speed or fixed beam and/or lower LOS velocity update rate (see [9], [10]).

The exception regarding the specific azimuth angle of 78° is investigated using Fig. 14. This figure shows the schematic of the range dependence of detectability for the cases of high and low detectability. Both cases can appear in one PPI scanning, for example, because of the aerosol distribution depending on the azimuth direction. In the experiments of this article, the lag-angle compensation could be performed just for one range. This causes a negative effect for the nearer ranges as shown in the figure. If the measurable range for the low detectability case is shorter than the target range, the lag-angle compensation makes the measurable range shorter. This theoretical investigation agrees with the experimental result for the abovementioned azimuth angle. In the future development, this problem can be prevented by using the adaptive optimization of the target range for the lag-angle compensation.

V. CONCLUSION

The experimental evidence of the decreased heterodyne-detection efficiency caused by the lag-angle was shown using our long-range wind sensing CDL technology with fast beam scanning. The recovery of the efficiency with the lag-angle compensation was also demonstrated. In the compensation, we applied the receiving beam alignment method synchronized with the beam scanner. The measurable range of 12 km with the fast beam scanning of 20°/s in the case of 8-Hz LOS update rate was demonstrated with the compensation. This demonstration on the recovery effect of the heterodyne-detection efficiency due to the lag-angle compensation will be the last big progress in the performance of the wind sensing CDL regarding the system efficiency. This progress makes the wind sensing CDL possible to satisfy all the requirements of long-range, real-time, and fast beam scanning measurement. This successful demonstration on the lag-angle compensation also implies the future possibility for the application to the spaceborne wind sensing CDL.

APPENDIX

The SLC night and day models have been used in the performance prediction in Sections IV-B and IV-C. The values are the following.

Night Model:

$$C_n^2(z) = \begin{cases} 8.40 \times 10^{-15} & z < 18.5 \\ 2.87 \times 10^{-12}/z^2 & 18.5 < z < 110 \\ 8.40 \times 10^{-15} & 110 < z < 1500 \\ 8.87 \times 10^{-7}/z^3 & 1500 < z < 7200 \\ 2.00 \times 10^{-16}/z^{0.5} & 7200 < z < 20\,000. \end{cases} \quad \text{(A1)}$$

Day Model:

$$C_n^2(z) = \begin{cases} 1.70 \times 10^{-14} & z < 18.5 \\ 3.13 \times 10^{-13} & 18.5 < z < 240 \\ 1.30 \times 10^{-15} & 240 < z < 880 \\ 8.87 \times 10^{-7}/z^3 & 880 < z < 7200 \\ 2.00 \times 10^{-16}/z^{0.5} & 7200 < z < 20\,000. \end{cases} \quad \text{(A2)}$$

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